MedForge AI | Research report

Research and educational demonstration only. Not medical diagnosis or real patient care. Use synthetic inputs only.

Model: Logistic regression | Positive-class probability: 0.127

A model score on a synthetic record; not an individual disease risk estimate.

Feature	Synthetic value	SHAP contribution
cp	3.0	-0.1415
ca	0.0	-0.1040
slope	2.0	+0.0681
thal	3.0	-0.0671
sex	1.0	+0.0426
restecg	0.0	-0.0351
exang	0.0	-0.0250
thalach	162.0	-0.0226
oldpeak	1.2	+0.0128
trestbps	128.0	-0.0101
chol	238.0	-0.0093
fbs	0.0	+0.0034
age	54.0	+0.0008

SHAP baseline 0.4145; baseline + contributions = probability. Units: probability points.

Held-out model evaluation

Logistic regression: ROC AUC 0.966, accuracy 0.885, recall 0.929; training CV AUC 0.902.

Random forest: ROC AUC 0.952, accuracy 0.869, recall 0.893; training CV AUC 0.896.

Gradient boosting: ROC AUC 0.962, accuracy 0.902, recall 0.929; training CV AUC 0.863.

Dataset: 303 records; train 242, test 61; random seed 42. Target: num > 0. Source: https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data

Limitations

Small historical referral cohort; not representative of contemporary populations.

Single held-out split; uncertainty is substantial with 61 test records.

No external validation, prospective study, clinical calibration or fairness certification.

SHAP describes model associations, not causes or treatment effects.